Smart Bulb Security Analysis

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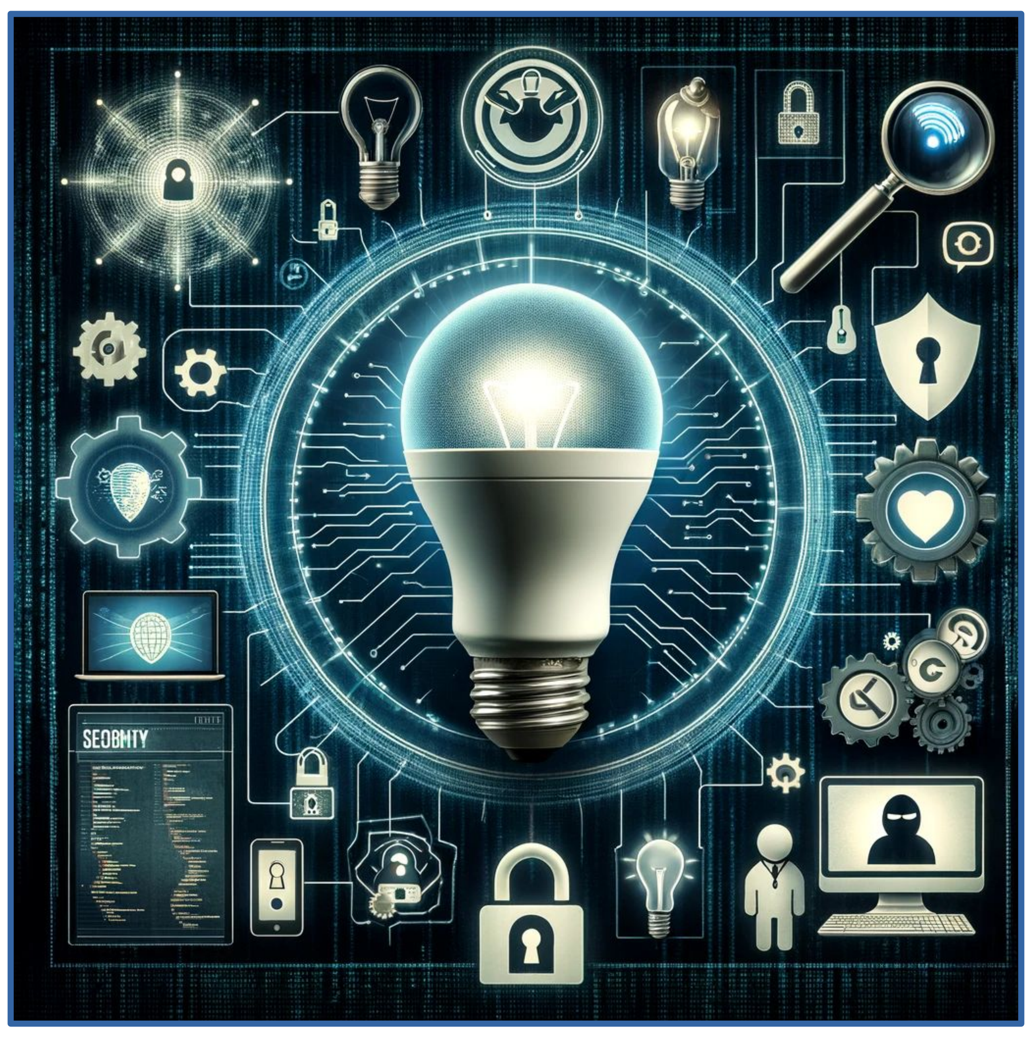
Problem Statement and Goals



Problem Statement:

- **Rise of Smart Home Devices:** Smart bulbs, integral to modern automation, pose increased cybersecurity risks.
- **Security Concerns:** Potential for unauthorized access, data breaches, and exploitation by malicious actors in smart bulbs.

Goals:

Identify Vulnerabilities: Examine security in smart bulbs (Cync, Merkury, Philips) using penetration testing, reverse engineering, and vulnerability scanning.
Assess Impact: Evaluate consequences of vulnerabilities in terms of unauthorized access, data leakage, and device damage.
Compare Security Posture: Analyze and contrast security features across different price ranges, linking cost to security efficacy.
Recommendations: Provide actionable suggestions for vulnerability mitigation and security enhancement for manufacturers and users.

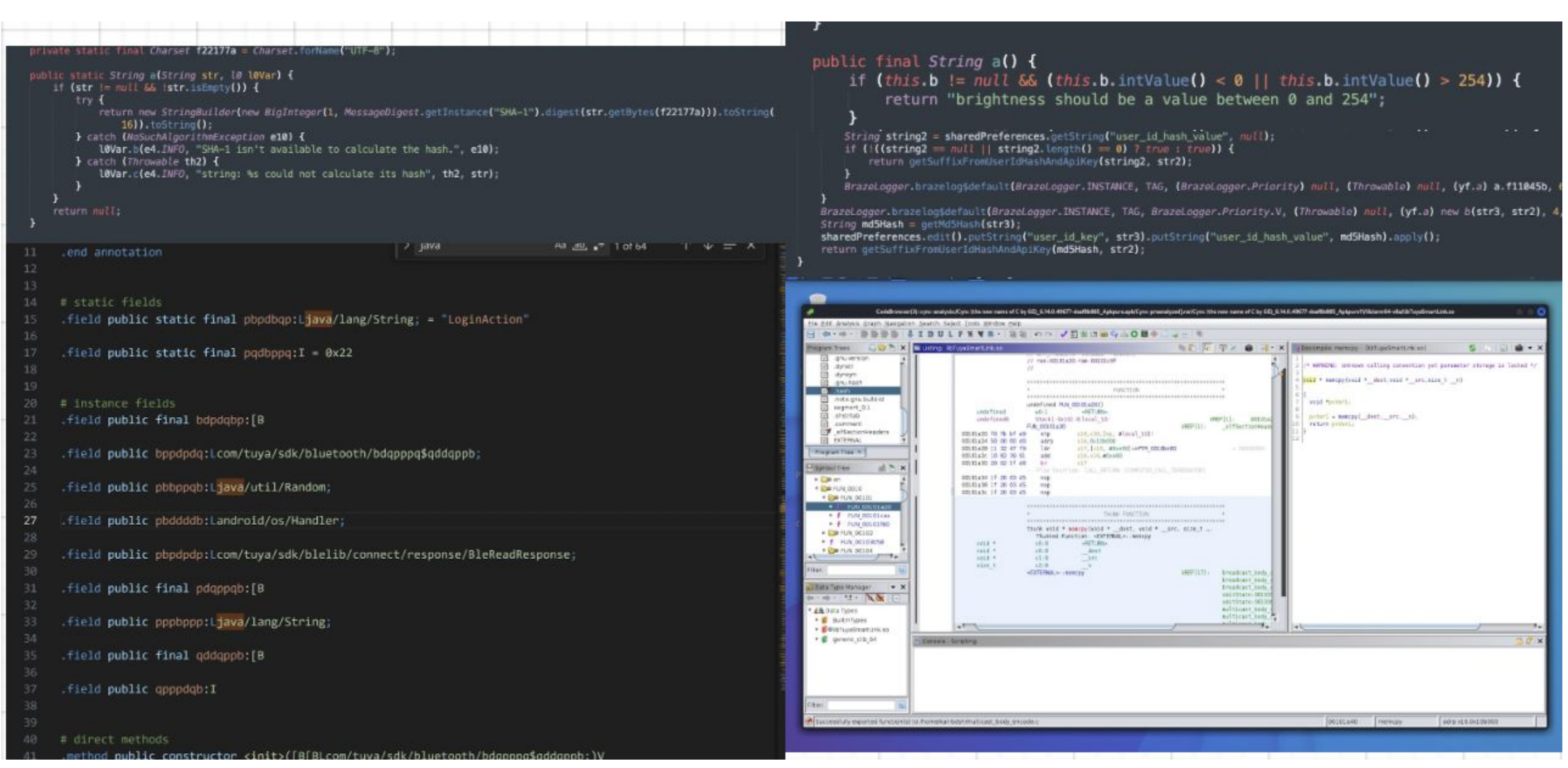
Approach

1. Merkury Smart Bulbs
- **Static Analysis:** Utilizing DeGuard and MobSF for deobfuscation and code analysis.
- **Dynamic Analysis Preparation:** Jailbreaking with Magisk for superuser access.
- **Network Analysis:** Employing BurpSuite and WireShark for traffic analysis on a controlled wireless network.

2. Cync by GE
- **Static Analysis:** Employing APKlab, MobSF, and Ghidra for deobfuscation and detailed code analysis.
- **Dynamic Analysis:** Using Wireshark and custom scripts for monitoring network traffic and runtime behavior.

3. Philips Hue
- **Static Analysis:** Utilizing apkdeguard, JADX, dex2jar, and JD-GUI for initial code analysis and deobfuscation.
- **MobSF Analysis:** Automated static analysis for vulnerability identification.
- **Dynamic Analysis:** Implementing Frida for script injection to test bulb brightness control.

Results



1. Merkury Smart Bulbs:
- **Permission and Manifest Analysis:** Identified excessive permissions and shared Activity-Alias leading to potential security risks.
- **Hardcoded Secrets:** Found exposed API keys, posing a threat to security.
- **Encryption and WebView Issues:** Discovered vulnerabilities in encryption and insecure WebView implementation.
- **Network Analysis:** Revealed hardware vulnerabilities (ESP32 chipset) and cloud infrastructure insights.

2. Cync by GE:
- **Weak Encryption and Cryptographic Algorithms:** Detected use of deprecated TLS/SSL versions and weak random number generators.
- **Unsafe Library Usage:** Found potential buffer overflow and integer overflow vulnerabilities.

3. Philips Hue:
- **Permission Issues:** Discovered discrepancies between app permissions and privacy policy.
- **Weak Hashing Algorithms:** Identified use of deprecated MD5 and SHA-1 hashing algorithms.
- **Brightness Adjustments:** Attempted testing of brightness control but faced challenges due to obfuscation.

Reference: OpenAI. "DALL-E." 2023, https://openai.com/dall-e.